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A 0.75-V Current Reference Based on Reverse Current Gain of BJT in Saturation Mode

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***Abstract*—This article introduces a current reference built upon a new reference basis derived from the reverse current gain (β_R) of a saturation-region bipolar junction transistor (BJT). By arranging two identical n-p-n BJTs in a self-cascode configuration, the proposed extraction mechanism establishes a millivolt-level reference voltage set by intrinsic bipolar device parameters. Unlike conventional BJT-based references anchored to the bandgap voltage (~1.2 V), the proposed basis reduces the required resistor by more than two orders of magnitude while maintaining a bounded process sensitivity through the logarithmic β_R dependence, a combination not simultaneously achieved by prior BJT- or CMOS-based approaches. Fabricated in a standard 0.18-μm CMOS technology with an active area of 0.025 mm^2, the proposed current reference generates 12.25 nA from a 0.75-V supply while consuming only 30 nW. With a single common trimming code applied across all chips, the prototype achieves a temperature coefficient (TC) of 277 ppm/°C over −40 °C to 125 °C and a line sensitivity (LS) of 0.34%/V over 0.75–2.0 V.**

***Index Terms*—Bipolar junction transistor (BJT), current reference, reverse current gain, subthreshold current, temperature coefficient (TC), ultra-low power (ULP).**

I. Introduction

PREDICTABLE on-chip current generation is fundamental to analog and mixed-signal systems, serving as the primary biasing source for core building blocks such as amplifiers, data converters, and oscillators. This requirement becomes particularly stringent in always-on Internet-of-Things (IoT) applications [1], [2], [3], [4], [5] where circuits must sustain robustness against process, voltage, and temperature (PVT) variations under severely constrained energy budgets. Regardless of topology, any current reference inherently operates by converting a well-defined *reference basis voltage* into a current through a resistive element. In bipolar junction transistor (BJT)-based designs, this basis is the bandgap voltage (V_{BG}); in CMOS-based designs, it is the threshold voltage (V_{TH}). In practice, this basis voltage is not directly accessible; it is embedded within a complementary-to-absolute-temperature (CTAT) voltage, such as the base-to-emitter voltage (V_{BE}) for BJTs or the gate-to-source voltage (V_{GS}) for CMOS. This CTAT voltage can be decomposed as $V_{CTAT}(T) = V_{basis} + A \times T$, where $A < 0$ is the first-order temperature coefficient (TC). Extracting V_{basis} from V_{CTAT} therefore requires the circuit to compensate for A across temperature. Consequently, the essential metrics, including power–area tradeoff and robustness of the reference current, are determined by the magnitude and the PVT stability of V_{basis} and A [6]. Table I summarizes the CTAT voltages and their corresponding reference bases for representative BJT- and CMOS-based current references.

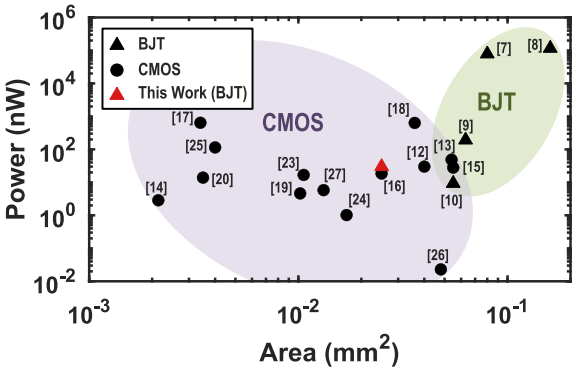


Fig. 1. Power versus area of the recent current references.

Traditional BJT-based references exploit the well-characterized temperature behavior of the V_{BE}, whose underlying basis is the silicon bandgap voltage ($V_{BG} \approx 1.2$ V). The logarithmic relationship between V_{BE} and the BJT saturation current inherently suppresses process-induced variations, providing strong immunity to process spreads without trimming. Recent works have further improved temperature stability through multi-point compensation techniques [7], [8], achieving excellent PVT robustness. However, all such approaches remain anchored to V_{BG}: generating a nanoampere-range current requires a resistor proportional to V_{BG}/I_{REF}, demanding multi-megaohm implementations that impose severe area overhead [7], [9]. This area penalty is evident in Fig. 1, where BJT-based designs cluster in the high-area region of the power–area space despite their excellent PVT robustness. Furthermore, the minimum supply voltage cannot be reduced below V_{BG},

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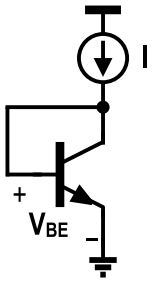
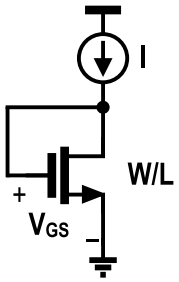
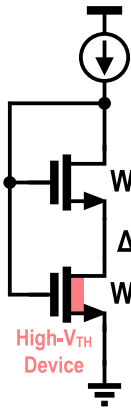
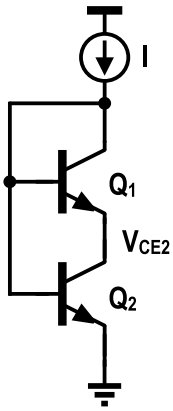
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TABLE I

COMPARISON OF HOST VOLTAGES AND REFERENCE BASES ACROSS BJT- AND CMOS-BASED CURRENT REFERENCES

CTAT/PTAT Voltage	V_{BE} (CTAT)	V_{GS} (CTAT)	ΔV_{GS} (CTAT/PTAT)	V_{CE2} (PTAT) [This Work]
Core Devices & Equation				
	$V_{BE} = V_{BG} - V_T \ln\left(\frac{I_{0,BJT}}{I}\right)$	$V_{GS} = V_{TH} - mV_T \ln\left(\frac{W}{L} \cdot \frac{I_{0,MOS}}{I}\right)$	$\Delta V_{GS} = \Delta V_{TH} - mV_T \ln\left(\frac{W_2/L_2}{W_1/L_1}\right)$	$V_{CE2} = V_{\beta R} + V_T \times \alpha$
Reference Basis (V_{basis})	V_{BG} (~1.2 V)	V_{TH} (300 ~ 500 mV)	ΔV_{TH} (< 350 mV)	$V_{\beta R}$ (~10 mV)
Slope (A)	-2 mV/°C	-1 mV/°C	Near-Zero	+0.04 mV/°C
Process Variation	Log	Linear	Linear + Uncorrelated	Log

restricting applicability in sub-1 V systems. In a single-BJT topology [10], the reference basis is retained at the bandgap voltage, while the resistor bias is reduced at the voltage-to-current (V–I) conversion stage by exploiting the drain-induced barrier lowering (DIBL) effect, so that a smaller resistance is required for the same current level. However, since this reduced resistance requirement relies on the DIBL effect, it introduces additional temperature dependence, resulting in a high untrimmed TC. The resistor bias can also be reduced by applying the V_{BE} difference of two BJTs, a proportional-to-absolute-temperature (PTAT) voltage, across a silicided resistor with a matching PTAT coefficient [11]. However, since this voltage has no temperature-independent component, the temperature stability relies on how closely the two TCs track over the range of interest, rather than on a reference basis of defined magnitude and process sensitivity.

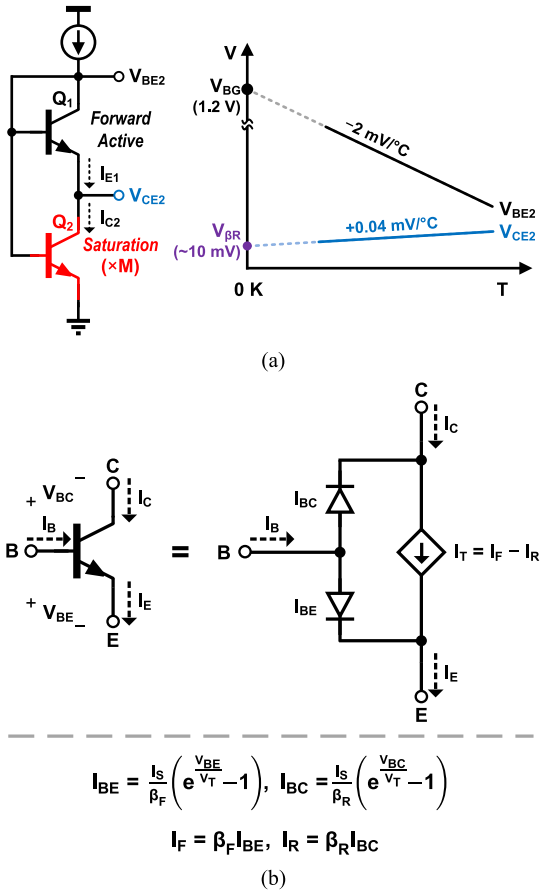
Alternatively, CMOS-based references replace V_{BG} with the threshold voltage V_{TH} or its difference ΔV_{TH} as the reference basis, offering a scalable magnitude that directly reduces the required resistor. Several designs have demonstrated nA-range operation with compact implementations [12], [13], [14], [15], [16], [17], [18], [19]. However, V_{TH} varies linearly with process-induced deviations in channel doping concentration and gate oxide thickness, making it considerably more sensitive to process spread across fabrication corners [20]. When multiple device types are involved, the uncorrelated nature of their threshold voltage variations across corners further amplifies this sensitivity [14], [15], [23]. Additionally, the higher order temperature dependence of carrier mobility and V_{TH} [21] introduces nonlinear curvature that degrades temperature linearity over a wide range. As a result, achieving competitive TCs in CMOS-based designs typically necessitates labor-intensive trimming [14], [22], [23], which inflates production cost and test time. Resistor-less implementations [23], [24], [25], [26], [27] reduce area further by replacing the resistor with appropriately biased transistors, but the exponential dependence of subthreshold current on V_{GS} renders the generated current highly susceptible to PVT variations.

This article addresses this fundamental limitation by introducing a new reference basis derived from the reverse current gain (β_R) of a saturation-region BJT, denoted $V_{\beta R}$. By exploiting the self-cascode interaction of two identical n-p-n BJTs, the proposed extraction mechanism establishes a millivolt-level reference voltage determined by bipolar device parameters, bypassing both the large magnitude of V_{BG} and the process sensitivity of V_{TH}. As analytically established in Sections II and III, $V_{\beta R}$ simultaneously achieves millivolt-level magnitude and process sensitivity between bandgap and threshold bases. Based on this principle, a low-voltage and power–area efficient current reference is implemented and experimentally validated in a standard 0.18-μm CMOS process.

The remainder of this article is organized as follows. Section II introduces the proposed β_R-based voltage extraction mechanism and analytically establishes $V_{\beta R}$ as a new reference basis, followed by a quantitative comparison against conventional reference bases. Section III describes the circuit implementation. Section IV presents measurement results and a comparison with the state of the art. Section V concludes this article.

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(a)

(b)

Fig. 2. (a) Voltage extraction from BJT self-cascode configuration and (b) transport model of n-p-n BJT.

II. Operation Principle and Characterization Analysis

A. Proposed β_R-Based Voltage Extraction

Fig. 2(a) illustrates the proposed reference basis extraction circuit, comprising two identical n-p-n BJTs (Q_1 and Q_2) arranged in a self-cascode configuration, where Q_2 is implemented as M parallel unit transistors identical to Q_1, yielding an effective emitter size ratio of 1:M. In this topology, the intermediate node voltage V_{CE2} is intrinsically clamped to a millivolt level, forcing Q_1 into the forward-active region while driving Q_2 into saturation. The transport model [28], a reformulation of the Ebers–Moll model [29], provides a unified description of BJT DC behavior across all operating regions. As depicted in Fig. 2(b), it represents the device by two p-n diodes and a single current-controlled current source characterized by the forward and reverse common-emitter current gains, β_F and β_R. Applying this model to the self-cascode topology, the emitter current of Q_1 and the collector current of Q_2 are expressed as

$$I_{E1} = I_S \left[\left(e^{\frac{V_{BE1}}{V_T}} - e^{\frac{V_{BC1}}{V_T}} \right) + \frac{1}{\beta_F} \left(e^{\frac{V_{BE1}}{V_T}} - 1 \right) \right] \tag{1}$$

$$I_{C2} = M \cdot I_S \left[\left(e^{\frac{V_{BE2}}{V_T}} - e^{\frac{V_{BC2}}{V_T}} \right) - \frac{1}{\beta_R} \left(e^{\frac{V_{BC2}}{V_T}} - 1 \right) \right] \tag{2}$$

where V_T is the thermal voltage and I_S is the BJT saturation current.

Two boundary conditions simplify (1) and (2). First, the diode-connected configuration of Q_1 (i.e., $V_{BC1} = 0$) reduces the reverse exponential term in (1) to unity. Second, the terminal constraint $V_{BE1} = V_{BC2} = V_{BE2} - V_{CE2} \gg V_T$ ensures that forward exponential terms dominate their reverse counterparts. Under these conditions, (1) and (2) reduce to

$$I_{E1} \approx I_S \cdot \left(1 + \frac{1}{\beta_F} \right) e^{\frac{V_{BE2} - V_{CE2}}{V_T}} \tag{3}$$

$$I_{C2} \approx M \cdot I_S \left[e^{\frac{V_{BE2}}{V_T}} - \left(1 + \frac{1}{\beta_R} \right) e^{\frac{V_{BE2} - V_{CE2}}{V_T}} \right]. \tag{4}$$

Enforced by the self-cascode topology, Kirchhoff's current law (KCL) requires $I_{E1} = I_{C2}$. Equating (3) and (4) and solving for V_{CE2} yields the extracted intermediate voltage

$$V_{CE2}(T) = V_T \ln \left(1 + \frac{1}{M} + \frac{1}{\beta_R} + \frac{1}{M\beta_F} \right). \tag{5}$$

In the reverse mode, the lightly doped collector assumes the role of the emitter, resulting in an inherently small β_R. For typical BJTs, $10 \le \beta_F \le 500$ and $0 < \beta_R \le 10$ (most often <1), i.e., $\beta_R \ll \beta_F$ [28]. In advanced technology nodes, β_F can become smaller than this typical range. We have nevertheless confirmed in simulation that $\beta_F/\beta_R \approx 100$ across the entire temperature range in the process used in this work and that the condition $\beta_R \ll \beta_F$ remains preserved even in a more advanced node. The $1/M\beta_F$ term in (5) is therefore negligible compared with the effect of $1/\beta_R$, and V_{CE2} reduces to

$$V_{CE2}(T) \approx V_T \ln \left(1 + \frac{1}{M} + \frac{1}{\beta_R} \right). \tag{6}$$

Equation (6) constitutes the central result of this work. V_{CE2} is governed not by the conventional forward-biased bandgap junction but by the reverse transport behavior of the saturated BJT. V_{CE2} is determined entirely by two intrinsic structural parameters: the emitter size ratio M and the reverse current gain β_R. Importantly, this dependence involves no absolute current levels or external components.

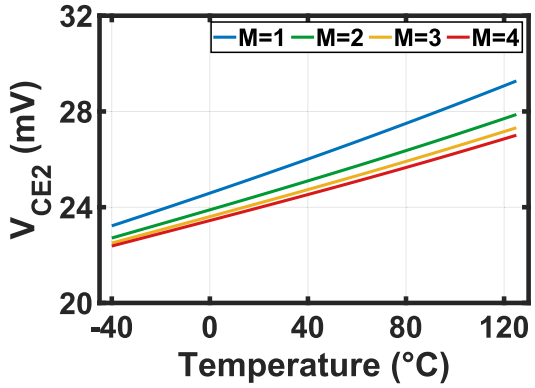
Since β_R reflects minority carrier transport in the reverse direction, its temperature dependence arises from the diffusivity asymmetry between the base and collector regions. Therefore, following the device-physics framework of [30], the reverse current gain can be expressed as

$$\beta_R \propto \frac{\mu_{minority,B}}{\mu_{minority,C}} \propto \left(\frac{T}{T_0} \right)^{\Delta n} \tag{7}$$

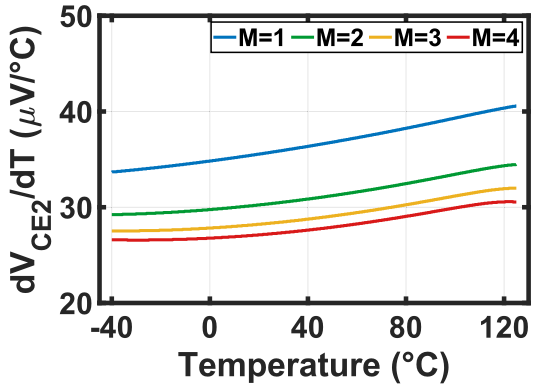
where $\mu_{minority,B}$ and $\mu_{minority,C}$ denote the minority carrier mobility in the base and collector regions, respectively—electrons in the P-type base and holes in the N-type collector. Both mobility and lifetime follow power-law temperature dependencies of the form $\propto T^{-n}$, where Δn is the net temperature exponent of β_R, dominated by the difference in power-law temperature exponents of minority-carrier mobility between the collector and base regions. Since minority carrier mobility in the lightly doped collector is dominated by phonon scattering, whereas in the more heavily doped base ionized-impurity scattering contributes an opposing temperature dependence, the collector exhibits a stronger temperature dependence than the base,

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(a)



(b)

Fig. 3. Simulation of temperature dependence of (a) V_{CE2} and (b) $\partial V_{CE2}/\partial T$ versus temperature with changing M from 1 to 4.

making Δn positive. While not a first-principles expression, this power-law model adequately describes the temperature trend of β_R used in the following analysis. With this model, the temperature dependence of $1/\beta_R$ is described as

$$\frac{1}{\beta_R(T)} = \frac{1}{\beta_R(T_0)}\left(\frac{T_0}{T}\right)^{\Delta n} \tag{8}$$

where T_0 is the nominal temperature. Substituting (8) into (6) and taking the first-order temperature derivative gives

$$\frac{\partial V_{CE2}}{\partial T} = \frac{k}{q}\left[\ln\left(1+\frac{1}{M}+\frac{1}{\beta_R(T)}\right) - \frac{\Delta n}{1+\left(1+\frac{1}{M}\right)\cdot\beta_R(T)}\right] \tag{9}$$

where k is Boltzmann's constant and q is the elementary charge. In (9), the first term originates from differentiating the V_T prefactor, and the second originates from the temperature dependence of $1/\beta_R$. Since the logarithmic argument exceeds unity, this term is positive, whereas the second term acts with the opposite sign. Fig. 3 illustrates the simulated V_{CE2} and $\partial V_{CE2}/\partial T$ across temperature while sweeping M from 1 to 4. The V_{CE2} plot exhibits the near-linear PTAT trend and the logarithmic attenuation of the slope with increasing M, as expected from (6). The $\partial V_{CE2}/\partial T$ plot increases monotonically and remains positive across the full temperature range in the process used, indicating that the first term dominates the second and thereby governs the PTAT slope of V_{CE2}. This monotonic, convex profile of the slope informed the selection of the composite resistor, whose higher order temperature characteristic is chosen to track it. Unlike conventional V_{BE} and V_{GS}, V_{CE2} is a PTAT voltage with a remarkably small intrinsic TC of ~ 40 μV/°C, which is 50 and 25 times smaller than the TCs of a BJT's base-to-emitter voltage and CMOS transistor's gate-to-source voltage, respectively.

Given the V_{CE2} trend in Fig. 3, where the slope varies only weakly with temperature, V_{CE2} can be linearly approximated using a Taylor expansion around T_0, expressed as

$$V_{CE2}(T) \approx \alpha \cdot V_T + V_{\beta R} \tag{10}$$

where the slope α and the temperature-independent intercept $V_{\beta R}$ are defined as

$$\alpha = \ln\left(1+\frac{1}{M}+\frac{1}{\beta_R(T_0)}\right) - \frac{\Delta n}{1+\left(1+\frac{1}{M}\right)\cdot\beta_R(T_0)} \tag{11}$$

$$V_{\beta R} = \frac{\Delta n \cdot V_T(T_0)}{1+\left(1+\frac{1}{M}\right)\cdot\beta_R(T_0)}. \tag{12}$$

Equations (11) and (12) establish that both α and $V_{\beta R}$ are determined solely by intrinsic bipolar device physics evaluated at T_0. In particular, (12) confirms that $V_{\beta R}$ is a physically well-defined, finite intercept determined by V_T, Δn, and β_R at T_0. The approximation error, evaluated analytically by comparing (10) with the exact expression in (6), remains within 1.5% across the entire temperature range of −40 °C to 125 °C.

B. Comparison With Conventional Reference Bases

The four reference bases introduced in Section I differ quantitatively in two key metrics: the voltage magnitude, which determines the required resistor for a target I_{REF}, and the normalized process sensitivity $\delta V_{CTAT}/V_{CTAT}$ (or $\delta V_{PTAT}/V_{PTAT}$), which represents the fundamental contribution of the reference basis to the overall current variation. To analytically establish the origin of these differences, the conventional device voltages are expressed in a unified form that explicitly separates the temperature-independent component from the temperature-dependent term as

$$V_{BE} = V_{BG} - V_T \ln\left(\frac{I_{0,BJT}}{I}\right) \tag{13}$$

$$V_{GS} = V_{TH} - mV_T \ln\left(\frac{W}{L}\cdot\frac{I_{0,MOS}}{I}\right) \tag{14}$$

$$\Delta V_{GS} = \Delta V_{TH} - mV_T \ln\left(\frac{W_2/L_2}{W_1/L_1}\right) \tag{15}$$

where $I_{0,BJT} = bT^{5/2}$ is a pre-exponential current coefficient of the BJT saturation current, with b being the proportionality factor [31]; and $I_{0,MOS} = \mu C_{OX}(m-1)V_T^2$ is a process-dependent parameter, with μ, C_{OX}, and m denoting the effective channel mobility, oxide capacitance, and subthreshold slope factor, respectively.

The temperature-independent intercepts in (13)–(15) directly set the reference basis voltages after TC reduction and hence the required resistor for a target V_{basis}/I_{REF}. From (12), $V_{\beta R}$ yields a nominal magnitude of ~8 mV—approximately 150× smaller than V_{BG}—reducing the required resistor to the kilohm range for typical nanoampere-level current generation,

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a reduction of more than two orders of magnitude relative to conventional bandgap-based BJT designs that bias the resistor with the full V_{BG}.

For the process sensitivity comparison, since $I_{REF} = V_{basis}/R$, the normalized sensitivity $\delta V_{CTAT}/V_{CTAT}$ (or $\delta V_{PTAT}/V_{PTAT}$) represents the fundamental contribution of the reference basis to the overall current variation, independently of resistor tolerance, and therefore serves as the figure of merit for comparing the intrinsic process sensitivity of each basis. For V_{BE}, the normalized sensitivity is obtained by taking the first-order partial derivative of (13) with respect to $I_{0,BJT}$

$$\frac{\delta V_{BE}}{V_{BE}} \approx \frac{V_T}{V_{BE}} \cdot \frac{\delta I_{0,BJT}}{I_{0,BJT}}. \tag{16}$$

The prefactor $V_T/V_{BE} \approx 0.05$, where $V_{BE} \approx 500$ mV represents a conservative minimum across the operating temperature range, strongly attenuates $I_{0,BJT}$ variation.

For V_{GS}, V_{TH} can vary linearly with channel doping N_A and gate oxide thickness t_{ox} across corners. Since the contribution of m and $I_{0,MOS}$ variation in (14) is negligible due to their weak process dependence and logarithmic compression relative to δV_{TH}, the first-order partial derivative of (14) simplifies to

$$\frac{\delta V_{GS}}{V_{GS}} \approx \frac{\delta V_{TH}}{V_{TH}}. \tag{17}$$

In a standard CMOS process, δV_{TH} across the SS-to-FF corner can be significant relative to V_{TH} itself, depending on device geometry and foundry process [32], yielding a normalized sensitivity considerably larger than that of V_{BE} in (16). For ΔV_{GS}, the two device types depend on distinct doping implants with statistically independent process distributions

$$\frac{\delta(\Delta V_{GS})}{\Delta V_{GS}} = \frac{\sqrt{\delta V_{TH,1}^2 + \delta V_{TH,2}^2}}{\Delta V_{TH}}. \tag{18}$$

The numerator exceeds δV_{TH} of a single device, while the denominator ΔV_{TH} is substantially smaller than V_{TH} itself, indicating that ΔV_{TH}-based references are inherently susceptible to elevated normalized process uncertainty relative to single-device V_{TH} approaches.

For the proposed V_{CE2}, the primary source of process variation is β_R. Under a process-induced shift $\delta(1/\beta_R)$, subtracting (6) from the perturbed expression and applying the first-order approximation

$$\delta V_{CE2} \approx \frac{V_T \cdot \delta\left(\frac{1}{\beta_R}\right)}{1 + \frac{1}{M} + \frac{1}{\beta_R}}. \tag{19}$$

Normalizing by (6)

$$\frac{\delta V_{CE2}}{V_{CE2}} = \frac{\left(\frac{1}{\beta_R}\right)}{\left(1 + \frac{1}{M} + \frac{1}{\beta_R}\right) \cdot \ln\left(1 + \frac{1}{M} + \frac{1}{\beta_R}\right)} \cdot \frac{\delta\left(\frac{1}{\beta_R}\right)}{\frac{1}{\beta_R}}. \tag{20}$$

The prefactor evaluates to $(1/\beta_R)/[(1 + 1/M + 1/\beta_R) \cdot \ln(1 + 1/M + 1/\beta_R)] \approx 0.37$ based on simulation, compared to $V_T/V_{BE} \approx 0.05$ in (16). Unlike V_{TH}- and ΔV_{TH}-based references where process variation maps linearly onto the CTAT voltage, the β_R variation is here attenuated by the prefactor arising from the logarithmic form of (6), yielding the reduced normalized sensitivity in (20) at the extraction node.

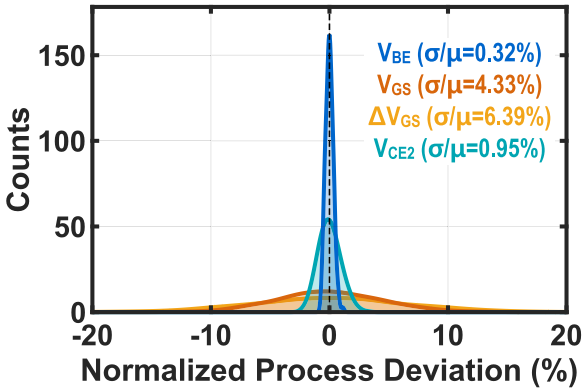


Fig. 4. Simulated process deviation distributions of V_{BE}, V_{GS}, ΔV_{GS}, and V_{CE2} at T = 30 °C, normalized to each mean value.

These analytical predictions are directly validated in Fig. 4, which presents the process deviation distributions of device voltages obtained from 500-run Monte Carlo simulations at T = 30 °C. Each circuit in Table I is configured under a common level of bias current (=10 nA), and the resulting distributions are normalized to their respective mean values to enable a fair comparison of intrinsic process sensitivity. The proposed BJT self-cascode topology extracts a millivolt-level reference basis $V_{\beta R}$, combining the reduced power–area overhead of low-voltage designs with a bounded process sensitivity at the extraction node. Its normalized sensitivity lies between that of conventional BJT bandgap and CMOS-based references, offering lower process sensitivity than the latter without matching the former.

Replacing the self-cascode BJT with a self-cascode CMOS would reduce this topology to a CMOS-stack reference such as [13] or [15]. In such realizations, the ΔV_{TH} basis equals V_{DS} of the lower transistor. Since the subthreshold current becomes drain-to-source voltage independent for $V_{DS} > 4V_T$, a requirement that tightens with temperature from 80 mV at −40 °C to 137 mV at 125 °C, the ~50 mV basis in [13] does not satisfy this condition, so the drain-to-source voltage factor does not fully vanish and the closed-form description begins to break down. This also forces a tradeoff between basis magnitude and corner robustness: enlarging the basis (~370 mV in [15]) admits a larger corner spread, while keeping it small, as in [13], leaves the V_{DS} constraint in force. The proposed self-cascode BJT avoids both limitations, extracting a millivolt-level basis from identical BJTs that retains a closed-form logarithmic model while keeping corner variation low. Moreover, because β_R is fixed by the process, the basis slope is predetermined, so the design separates into two independent choices: the input-pair ratio for slope cancellation and the resistance for the target current, whereas in a self-cascode CMOS, the same transistor dimensions set the threshold voltage, its TC, and the basis at once.

III. Circuit Description

A. Overall Architecture

The proposed current reference implements the β_R-based extraction principle established in Section II through a three-stage signal chain: 1) V_{CE2} generation; 2) voltage mirroring

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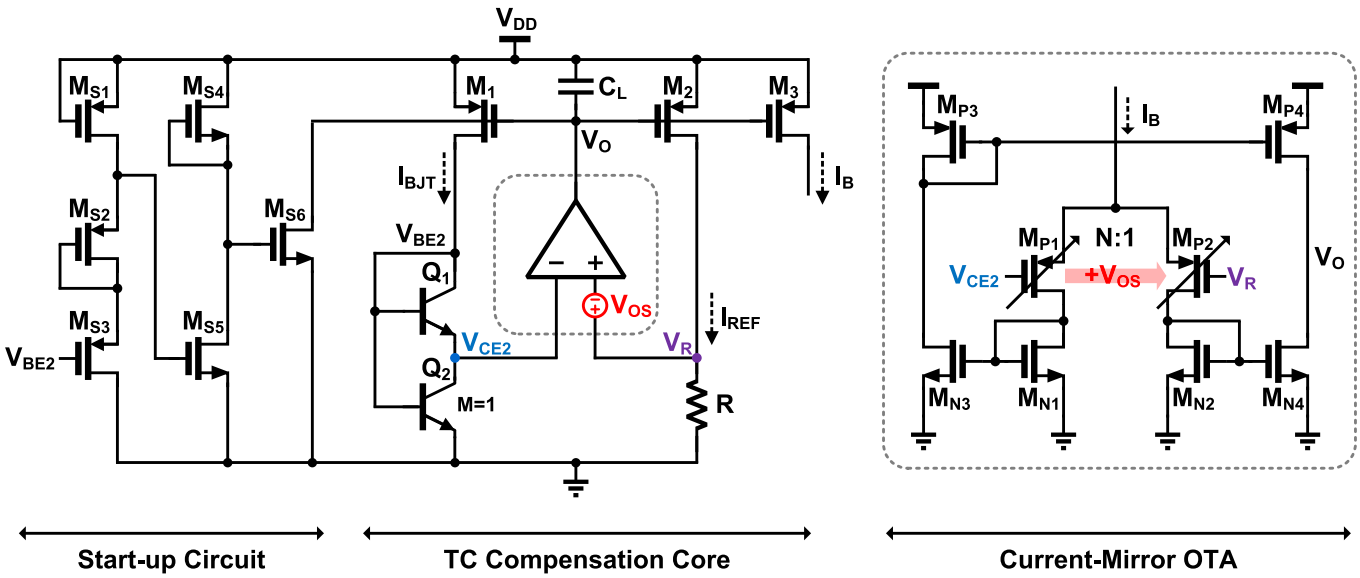


Fig. 5. Overall schematic of the proposed current reference circuit.

and temperature compensation to produce V_R; and 3) V–I conversion to generate I_{REF}. The overall schematic, including the start-up circuit, is shown in Fig. 5.

In the first stage, the self-cascode BJT pair (Q_1, Q_2) in the left branch physically realizes the extraction mechanism of (6) and sets V_{CE2} at the intermediate node. This node exhibits a tightly bounded spread across process corners by the logarithmic suppression of β_R variation established in (20). While a larger M improves the temperature linearity of V_{CE2}, as indicated by (9), it does so at the expense of increased junction leakage, larger silicon area, and degraded process sensitivity. For this prototype of the β_R-based current reference, M is therefore set to 1.

In the second stage, a negative feedback loop enforces the extracted V_{CE2} onto the right branch. The loop is formed by a three-current-mirror operational transconductance amplifier (OTA), which drives the right branch until $V_R = V_{CE2} + V_{OS}$, where V_{OS} is an intentional input offset. This OTA topology minimizes the number of stacked transistors, preserving critical voltage headroom for subthreshold operation. The OTA adopts a self-biasing scheme wherein its bias current I_B is directly mirrored from I_{REF}, ensuring that the total supply current I_{DD} remains strictly proportional to I_{REF} across all operating conditions.

The temperature compensation strategy exploits the linear temperature decomposition established in (10). V_{CE2} exhibits a PTAT slope of αV_T with a temperature-independent intercept $V_{\beta R}$. To compensate for the TC of V_{CE2}, V_{OS} is implemented by skewing the size ratio of the OTA input pair (M_{P1}, M_{P2}). Operating in the subthreshold region, this deliberate mismatch induces a CTAT offset of $-mV_T \ln(N)$ [33]. By selecting N such that $mV_T \ln(N) = \alpha V_T$, this CTAT component cancels the first-order PTAT slope of V_{CE2} established in (10).

In the third stage, the temperature-independent $V_R = V_{\beta R}$ is converted into I_{REF} through the composite resistor R. While the voltage applied across R is temperature-independent, the physical resistor itself exhibits temperature variations. To counteract this, R (=0.66 MΩ) is implemented as a composite structure combining a PTAT P+ diffusion resistor R_{PTAT} and a CTAT high-resistivity P-type polysilicon resistor R_{CTAT}, such that their opposing TCs cancel over the operating range. The resulting reference current is

$$I_{REF} = \frac{V_{CE2} + V_{OS}}{R} = \frac{V_T \ln\left(1 + \frac{1}{M} + \frac{1}{\beta_R}\right) - mV_T \ln(N)}{R} \approx \frac{(\alpha V_T + V_{\beta R}) - mV_T \ln(N)}{R} = \frac{V_{\beta R}}{R}. \quad (21)$$

In (21), the linearized form $V_{\beta R}/R$ is used to infer the approximate magnitude of the reference current after slope cancellation, whereas the β_R dependence of I_{REF} is more accurately described by the logarithmic form in the numerator. Consistent with the process sensitivity analysis in Section II-B, which is derived from the logarithmic expression (6) rather than its linearized approximation, the process dependence is interpreted from this logarithmic form. Table II lists the component parameters of the proposed implementation in Fig. 5.

The minimum supply voltage, $V_{DD,Min}$, is constrained by the stacked voltage headroom of the reference generation branch.

TABLE II
DESIGN PARAMETERS OF THE PROPOSED REFERENCE

Symbol	Size	Symbol	Size
M_1	8 µm / 5 µm	M_{S1}	2.2 µm / 2 µm
M_2	5 µm / 5 µm	M_{S2}	0.22 µm / 2 µm
M_3	8 µm / 20 µm	M_{S3}	0.22 µm / 20 µm
M_{P1}	35 µm / 20 µm	M_{S4}	0.5 µm / 0.5 µm
M_{P2}	23 µm / 20 µm	M_{S5}	0.22 µm / 20 µm
M_{P3}	16 µm / 20 µm	M_{S6}	
M_{P4}		Q_1	17 µm / 17 µm
M_{N1}	12 µm / 20 µm	Q_2	17 µm / 17 µm
M_{N2}		C_L	4.38 pF
M_{N3}	8 µm / 20 µm	R_{PTAT}	0.26 MΩ
M_{N4}		R_{CTAT}	0.4 MΩ

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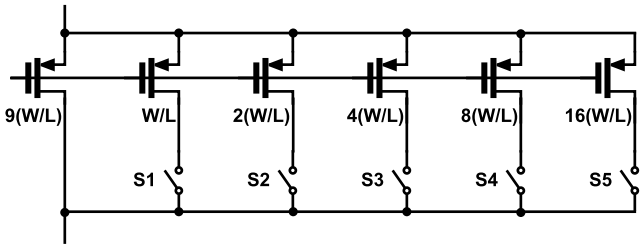


Fig. 6. Schematic of the trimming circuit used in the input pair of the OTA.

As depicted in Fig. 5, the critical path consists of the source-to-drain voltage of the current mirror PMOS M_1 ($V_{SD,M1}$) and the base-to-emitter voltage of Q_2 (V_{BE2}), giving

$$V_{DD} \geq V_{BE2} + V_{SD,M1}. \tag{22}$$

Since M_1 operates in the subthreshold region, the feedback loop regulates its V_{GS} to enforce current balance, allowing $V_{SD,M1}$ to remain small across operating conditions. Consequently, $V_{DD,Min}$ is predominantly determined by V_{BE2}. Since V_{BE2} is CTAT, it increases toward lower temperatures, so the lower end of the operating temperature range becomes the worst case; from (22), this yields $V_{DD,Min} \approx 0.75$ V—well below the V_{BG}-imposed floor of conventional BJT-based designs—making it directly compatible with sub-1 V systems.

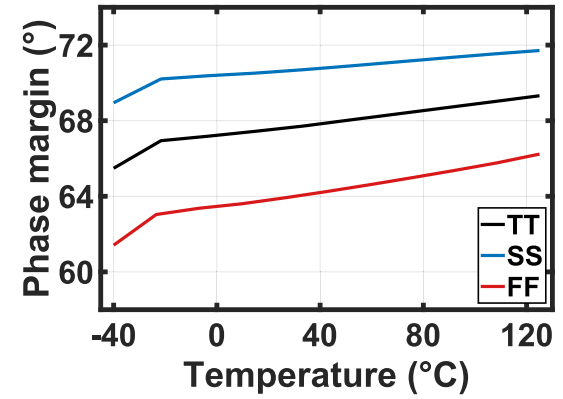
B. Trimming Circuit

Trimming approaches in the literature can be broadly classified into two categories: individual-code trimming [10], [12], [14], [22], [23] in which each chip is independently optimized to minimize its specific TC or output deviation, and common-code trimming in which a single code is uniformly applied to all chips [7], [8], [13], [15]. A common code can be predetermined from design and simulation steps or selected from batch-level characterization of measured samples, the latter being referred to as batch trimming. Individual-code trimming achieves the best per-chip accuracy but incurs significant test time and production cost, whereas common-code trimming sacrifices per-chip optimization in exchange for reduced calibration overhead.

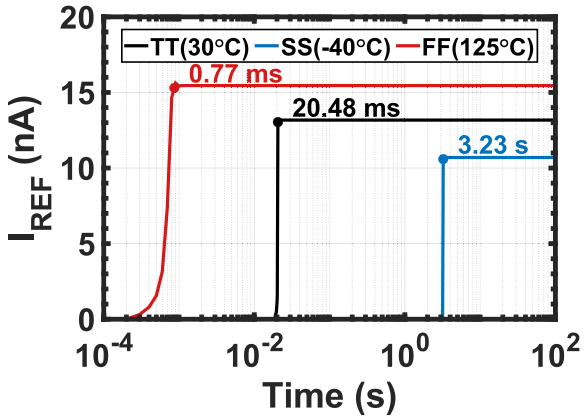
The proposed circuit incorporates a 5-bit binary-weighted trimming network in each branch of the OTA input pair (M_{P1}, M_{P2}), as shown in Fig. 6. By independently adjusting the numerator and denominator of the sizing ratio N, this trimming circuit provides fine control over V_{OS} and hence the degree of PTAT slope cancellation in (21), reaching ratios that adjusting a single side could not provide. Since V_{OS} is set by $\ln(N)$, a step in the ratio yields a smaller step in the offset. In this work, a common trimming code is applied to compensate for the overall TC variation arising from global process skews in the transistors and the composite resistor. This common code sets the OTA input-pair ratio to $N = 35/23$, determined by the batch trimming described in Section IV.

C. Stability and Statistical Analysis

The self-biased architecture inherently forms a positive feedback loop through the current mirror, which must coexist stably with the negative feedback enforced by the OTA. To ensure loop stability, a load capacitor C_L is added to introduce a dominant pole across PVT conditions. Additionally, a start-up circuit is required to escape the parasitic zero-bias operating point upon power-on [15] and [27]; in this design, M_{S1}–M_{S6} fulfill this role, becoming inactive once normal operation is established and contributing negligible leakage to the main loop. As shown in Fig. 7(a), the simulated phase margin remains above 60° across all PVT conditions, confirming stable operation of the self-biased architecture. Fig. 7(b) shows the simulated start-up waveforms in the typical (TT, 30 °C), fastest (FF, 125 °C), and slowest (SS, −40 °C) conditions. The nominal 99% start-up time is 20.48 ms in the typical condition, ranging from 0.77 ms (FF, 125 °C) to 3.23 s (SS, −40 °C).



(a)



(b)

Fig. 7. Simulated (a) phase margin versus temperature for the TT, SS, and FF corners and (b) 99% start-up waveform at the typical (TT, 30 °C), slowest (SS, −40 °C), and fastest (FF, 125 °C) conditions.

Fig. 8 presents the simulated I_{REF} across global process corners, where the transistors (TRs, including BJT and CMOS) and the composite resistor (R) are independently swept across TT, SS, and FF corners. As shown in the plot, variations in the TR corner primarily affect the temperature behavior of I_{REF}, whereas the resistor corner predominantly shifts DC magnitude of I_{REF}. The observed TC variation across TR corners motivates the common-code trimming strategy, which compensates for the global process skews shared across all chips with a single common code.

Fig. 9 illustrates the statistical robustness of the proposed architecture through 1000-run Monte Carlo simulations of

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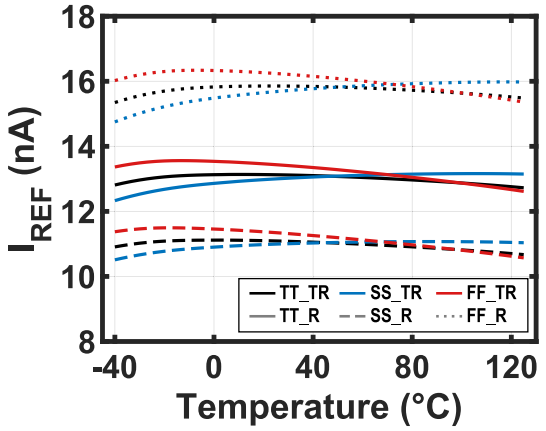


Fig. 8. Simulated I_{REF} versus temperature under independent TR and resistor process corner (TT, SS, and FF) sweeps.

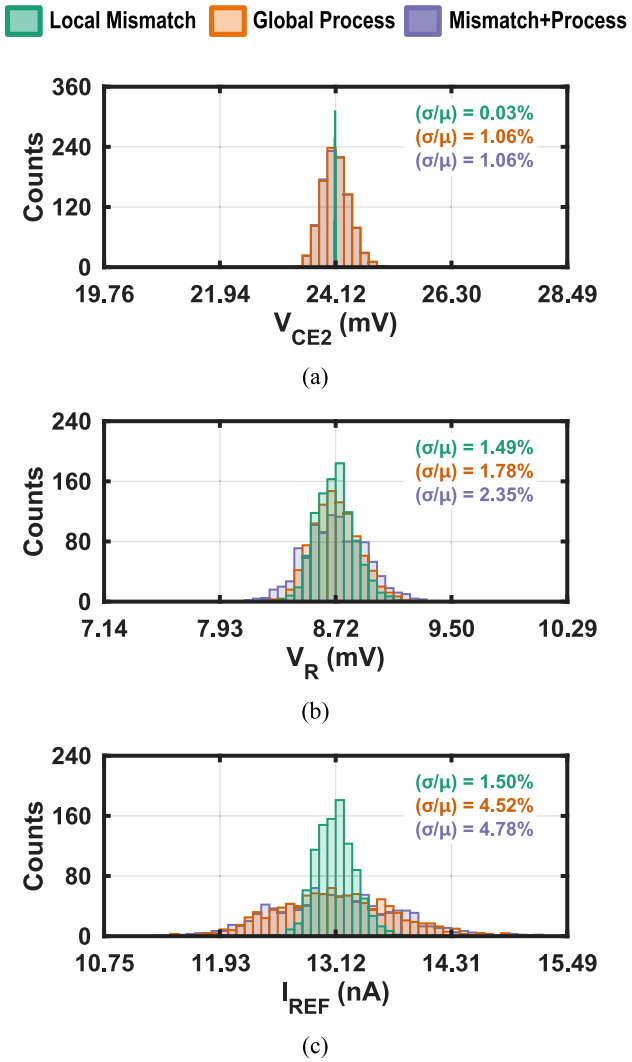


Fig. 9. Monte Carlo simulations with 1000 runs for (a) V_{CE2}, (b) V_R, and (c) I_{REF} at V_{DD} = 0.75 V and T = 30 °C.

V_{CE2}, V_R, and I_{REF}, decomposed into local mismatch, global process variation, and their combination. The effect of variance accumulation across the steps is visualized by plotting the histograms with bin widths normalized to the mean and standard deviation of I_{REF}. At the extraction node, V_{CE2} remains

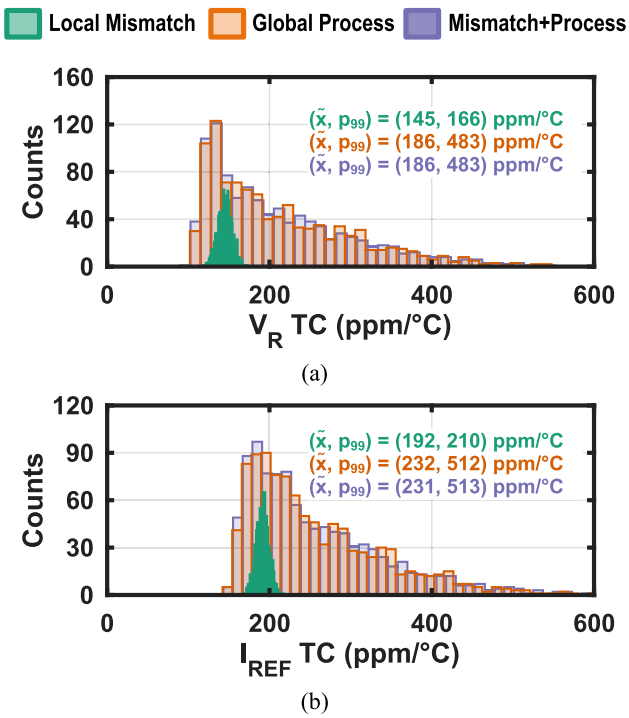


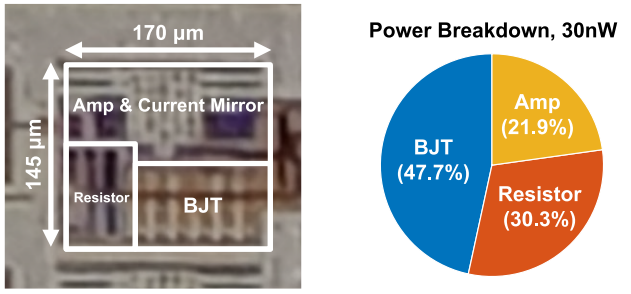
Fig. 10. Monte Carlo simulations with 1000 runs for TC of (a) V_R and (b) I_{REF} at V_{DD} = 0.75 V across a temperature range of −40 °C to 125 °C.

tightly bounded at 1.06%, consistent with the prefactor-based attenuation of β_R variation in (20). As the signal propagates to V_R, the local mismatch of the OTA input pair rises to 1.49%, becoming comparable to the global process variation (1.78%); the emergence of this local mismatch is thus the dominant new mechanism that broadens the combined V_R spread from 1.06% to 2.35%. Finally, during V–I conversion, the uncorrelated global process variation of R dominates (4.52%), far exceeding the 1.50% mismatch contribution and yielding a combined I_{REF} spread of 4.78%. The residual I_{REF} variation therefore originates primarily from the resistor process tolerance inherent to the V–I conversion—as is common to resistor-based references—rather than from the β_R extraction, whose basis-stage contribution remains at the ~1% level.

Fig. 10 presents the Monte Carlo distributions of the TC of V_R and I_{REF} over −40 °C to 125 °C. As these TC distributions are non-Gaussian and right-skewed, we report the median ($\tilde{x}$) and the 99th percentile (p_{99}) instead of the mean and standard deviation, following [14], [23], so as to characterize the worst case behavior rather than assume a symmetric distribution. For both quantities, local mismatch yields a narrow distribution whose $\tilde{x}$ and p_{99} nearly coincide [($\tilde{x}$, p_{99}) = (145, 166) ppm/°C for V_R and (192, 210) ppm/°C for I_{REF}], whereas global process variation introduces a pronounced tail, with p_{99} of 483 and 512 ppm/°C against medians of 186 and 232 ppm/°C, respectively. Since the combined distribution nearly coincides with the process-only case for both V_R and I_{REF}, the TC spread is governed by global process variation rather than local mismatch. Crucially, V_R—set by the feedback loop to $V_{CE2} + V_{OS}$ independently of R—already exhibits nearly the full TC spread (p_{99} = 483 ppm/°C), while the subsequent V–I conversion only shifts the median upward (186 → 232 ppm/°C) without

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(a) (b)

Fig. 11. (a) Die micrograph and (b) power breakdown.

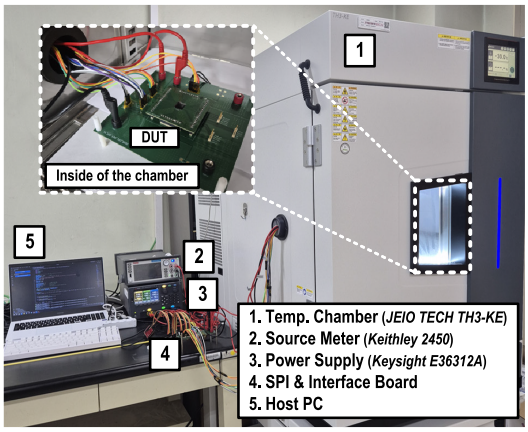


Fig. 12. Measurement setup.

broadening the tail (483 → 512 ppm/°C). This isolates the corner-dependent detuning of the slope cancellation between the PTAT slope of V_{CE2} (set by β_R) and the CTAT offset, which is set by the CMOS input pair—rather than the composite resistor, whose TC is engineered to be low—as the dominant source of TC variation.

IV. Measurement Results

The proposed current reference was fabricated in a standard 0.18-μm CMOS technology. Fig. 11(a) shows the die micrograph with an active area of 0.025 mm^2, and Fig. 11(b) details the measured power breakdown at the nominal 0.75-V supply at room temperature. The total power consumption of 30 nW is distributed among the BJT core (46.6%), the amplifier and current mirror (22.8%), and the composite resistor (30.6%). A total of ten samples were measured using the setup shown in Fig. 12. The device under test (DUT) is placed in a temperature chamber, with its supply provided by a power supply and the trimming code delivered from a microcontroller unit (MCU) via SPI. The entire measurement was controlled by a host PC over USB and GPIB. The output current $I_{OUT} = 32 \times I_{REF}$ is measured by a source meter Keithley 2450. The leakage of the electrostatic discharge (ESD) protection diodes in the chip pad increases to 38 pA at 125°C, adding less than 1 ppm/°C to the TC of the measured I_{REF}. The effect of the ESD leakage is therefore negligible. Meanwhile, the on-chip current mirror generating I_{OUT} increases the σ/μ by about 0.6% relative to the intrinsic I_{REF}.

Since directly probing the V_{CE2} node may affect the low-power operation and internal loop stability, V_R at the opposite

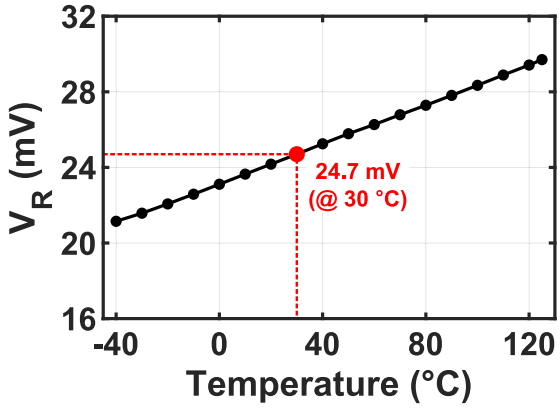


Fig. 13. Measured V_R at the zero-offset state ($N = 1$, $V_{OS} = 0$).

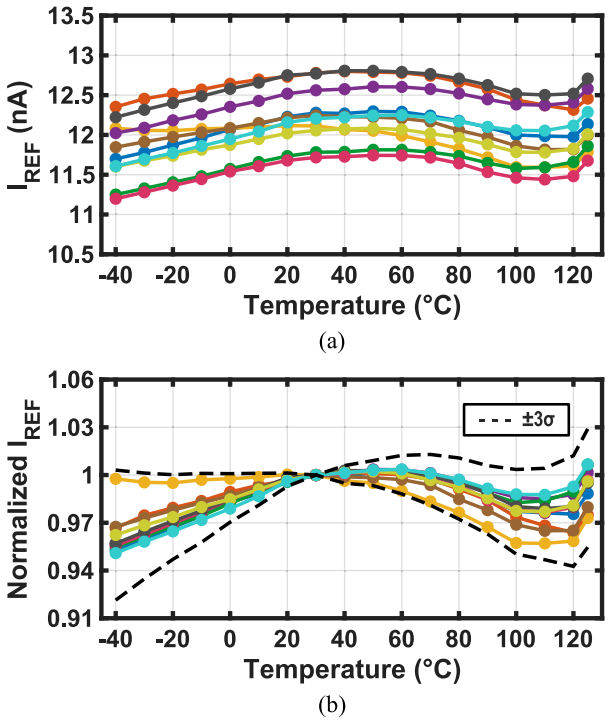


Fig. 14. Measured temperature dependence of (a) I_{REF} and (b) normalized I_{REF} from ten chips across a temperature range of −40 °C to 125 °C at V_{DD} = 0.75 V.

input node of the OTA is measured directly, with the trimming circuit configured to its zero-offset state ($N = 1$) so that $V_{OS} = 0$ and $V_R \approx V_{CE2}$. As shown in Fig. 13, the measured V_R at 30 °C is approximately 24.7 mV, averaged across the measured samples to reduce the influence of random mismatch. Across −40 °C to 125 °C, V_R varies over about 21–30 mV, exhibiting the near-linear PTAT slope originating from the β_R-based extraction of the BJT. While this measured V_R is an indirect indicator rather than the exact V_{CE2}, it is consistent with the expected magnitude and PTAT trend of V_{CE2}.

Fig. 14(a) and (b) plots the absolute and normalized temperature dependence of I_{REF} from −40 °C to 125 °C across all measured samples. The measurement over temperature was performed at $V_{DD} = 0.75$ V to verify that a stable I_{REF} is obtained even under minimum power operation. From batch-level characterization, combinations of the numerator and

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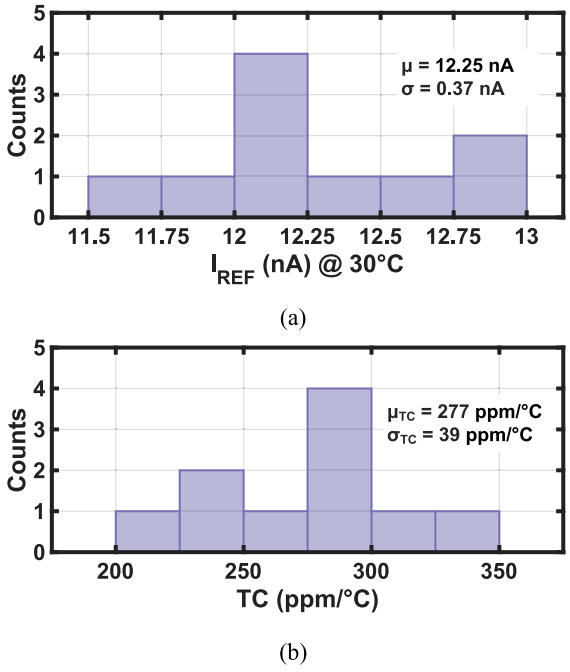


Fig. 15. Measured distributions of (a) I_{REF} at 30 °C and (b) TC.

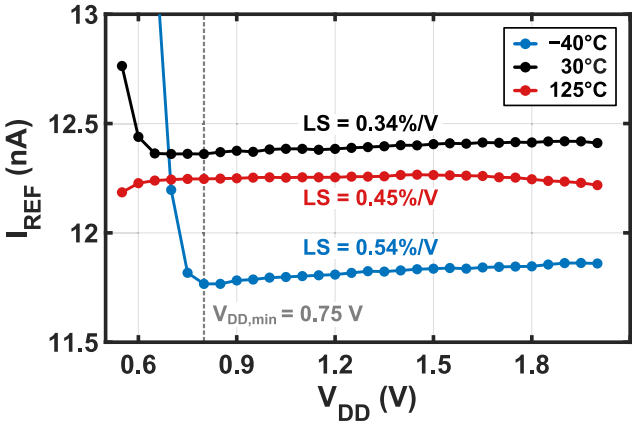


Fig. 16. Measured LS of I_{REF} at different temperatures.

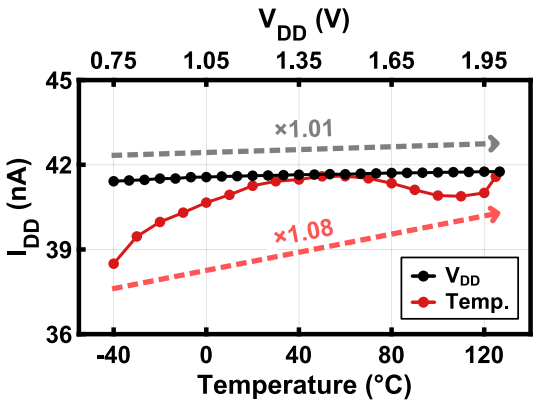


Fig. 17. Measured current consumption versus supply voltage and temperature.

denominator covering N from 1.50 to 1.56 were measured on all ten samples over the same temperature range. The combination giving the lowest batch-average TC, rather than the best TC of any individual sample, was $N = 35/23$, which was applied uniformly to every sample. At high temperature, the parasitic junction leakage becomes non-negligible in this nanoampere-level design, causing I_{REF} to rise slightly. The normalized profiles reveal that all samples share a consistent systematic curvature, showing that the dominant temperature behavior originates from the extraction mechanism itself rather than chip-to-chip random variation.

Fig. 15(a) and (b) presents the statistical distributions of I_{REF} at 30 °C and TC across the ten measured samples, respectively. The average I_{REF} at 30 °C is 12.25 nA with a standard deviation of 0.37 nA, which includes the variation added by the output current mirror. The measured TC records an average of 277 ppm/°C with a standard deviation of 39 ppm/°C. The residual chip-to-chip variation in this common-code design is predominantly governed by the local V_{TH} mismatch of the OTA input pair and the global process skew of the composite resistor. This observation is consistent with the Monte Carlo analysis in Section III. It should be noted that all ten samples were fabricated on a single wafer run; characterization across different fabrication batches was not performed in this study. Consequently, the reported σ/μ of 2.99% reflects chip-to-chip variation within the same batch rather than full process spread. Validation across multiple fabrication batches is left as future work.

Fig. 16 shows the measured line sensitivity (LS) at −40 °C, 30 °C, and 125 °C. In addition to the nominal room-temperature condition, both ends of the operating temperature range were examined: the lower end as the worst case for supply headroom identified in (22), and the upper end to confirm robustness at high temperature. The onset of insufficient headroom shifts to a higher V_{DD} as the temperature decreases, following the CTAT behavior of V_{BE2} in (22) and confirming the overall $V_{DD,Min}$ of 0.75 V. Over a supply range of 0.75–2.0 V, the average LS is 0.54%/V at −40 °C, 0.34%/V at 30 °C, and 0.45%/V at 125 °C.

Because scaled replicas of I_{REF} are distributed to both the core generation branch and the self-biased amplifier, the current consumption of the system remains strictly defined as a proportional function of I_{REF}. The measured total current varies by only 1.08× across the full temperature range and 1.01× across the operational supply voltage range, demonstrating the effectiveness of the self-biasing scheme in stabilizing power consumption under varying operating conditions (Fig. 17).

Directly measuring the noise of a nanoampere-level current is inherently difficult and further constrained by the measurement setup, so the characterization was performed after the output current mirror, where the reference current is scaled up 32 times. Fig. 18(a) shows the measured noise spectrum of I_{OUT}, and the 20 000-sample histogram in Fig. 18(b) yields an integrated noise of 265 $\mathrm{pA_{rms}}$ from 0.01 to 10 Hz. This prototype does not include a dedicated noise mitigation technique, as noise optimization was not among its primary objectives.

Table III summarizes the overall performance of the proposed β_R-based current reference and compares it with recently reported state-of-the-art low-power current references. Since the absolute power of a current reference scales with I_{REF} and

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TABLE III
PERFORMANCE SUMMARY AND COMPARISON

	This Work	[10] ISSCC'17	[7] JSSC'24	[8] ISSCC'26	[9] TCASII'21	[22] JSSC'20	[23] JSSC'24	[14] JSSC'25	[19] ESSERC'24	[13] TCASII'20	[12] TCASI'23
Technology (nm)	**180**	180	180	65	180	180	110	22	180	180	130
Type	**BJT**	BJT				CMOS					
Reference Basis	**$V_{\beta R}$**	V_{BG}	V_{BG}	V_{BG}	V_{BG}	ΔV_{TH}	ΔV_{TH}	ΔV_{TH}	ΔV_{TH}	ΔV_{TH}	V_{TH}
Minimum V_{DD} (V)	**0.75**	1.4	1.3	1.5	2	1.5	0.8	0.75	0.5	0.8	0.95
I_{REF} (A)	**12.25n**	6.64n	10.34µ	18.25µ	51n	1n	2.3n	1.5n	4.2n	11.6n	1.85n
Power Consumption (W)	**30n**	9.3n	78.3µ	116µ	192n	14n[a]	16.8n	2.87n	4.62n	48.64n	30n
Area (mm^2)	**0.025**	0.055	0.08	0.16	0.063	0.332	0.0106	0.00214	0.0102	0.054	0.04
V–I Converter	**Resistor**	Resistor	Resistor	Resistor	Resistor	Resistor	Transistor	Resistor	Resistor	Resistor	Resistor
Resistor Size (MΩ)	**0.66**	1.7	6.57	0.22	60.12	X[b]	X	N/A	1	3.1	20
Temperature Range (°C)	**-40 ~ 125**	0 ~ 110	-20 ~ 125	-40 ~ 125	-45 ~ 125	-20 ~ 80	-40 ~ 85	-40 ~ 85	-40 ~ 125	-40 ~ 120	-40 ~ 120
Trimming Code	**Common (Batch)**	Individual (N/A[c])	Common (Batch)	Common (Batch)	Individual (1-point)	Individual (1-point)	Individual (2-point)	Individual (2-point)	Individual (1-point)	Common[g] (Batch)	Individual (1-point)
$TC_{(Common)}$ (ppm/°C)	**277**	680	11.4	9.1	-	-	530	168	297	169	-
$TC_{(Individual)}$ (ppm/°C)	**-**	283	-	-	89	265	176	89	292	-	822
Measured $(\sigma/\mu)_{(Common)}$ (%)	**2.99**	-	0.54[d]	0.44[d]	-	-	2.11	4.1	7.62	4.3	15
Measured $(\sigma/\mu)_{(Individual)}$ (%)	**-**	1.19	-	-	1.15	0.25	3.47	0.61	-	-	10
Simulated $(\sigma/\mu)_{Process}$ (%)	**4.52**	4.07	N/A	N/A	N/A	N/A	6	10.48	5.4	4.65	N/A
Line Sensitivity (%/V)	**0.34**	1.16	0.036	0.16	1.76	1.4	2.23	0.51	0.27	1.08	4
# of Samples	**10**	10	45	20	30	10	11	11	10	10	10
Normalized FoM_1[e]	**1**	1.26	9.89	16.61	3.87	75.92	1.26	0.07	0.18	3.7	10.59
Normalized FoM_2[e]	**1**	4.63	0.46	0.55	1.21[f]	119.84[f]	3.19	0.05	0.2	2.33	32.42[f]

[a] 2.5-min cycle calibration required. [b] Switched capacitor $R_{eq} \approx 1/(f \cdot C)$. [c] Trimming method is not reported. [d] Measured variation from 5 corners.
[e] FoM_1 and FoM_2 are normalized to those of the proposed work. [f] Calculated with the individual trimming TC. [g] TC and power/LS use different trimming codes.

is therefore not meaningful on its own, the power is normalized to I_{REF}, and a first figure of merit (FoM) is defined as the product of this normalized power and the silicon area

$$\begin{aligned} \mathrm{FoM}_1 &= Area \times \left(\frac{Power}{1\mathrm{V} \times I_{REF}} \right) \\ &= Area \times \left(\frac{V_{DD}}{1\mathrm{V}} \right) \times \left(\frac{I_{DD}}{I_{REF}} \right). \end{aligned} \tag{23}$$

The normalized power thus reduces to the product of the supply voltage and the current overhead factor (I_{DD}/I_{REF}), and FoM_1 expresses the overall design cost of the reference along three axes—silicon area, supply voltage, and current overhead—with a lower value indicating a more resource-efficient design. These axes are not independent, since lowering the power requires a lower I_{REF} and hence a larger resistance and area; the product therefore reflects the cost of this tradeoff. As a fair normalization of silicon area to the technology node is not straightforward for analog circuits [14], [23], the areas in Table III are compared as reported.

FoM_1 quantifies what a design costs but not what it achieves, since the temperature stability that the reference ultimately provides is not represented. A second FoM is therefore defined by weighting FoM_1 with the TC referred to the characterized temperature range

$$\mathrm{FoM}_2 = \mathrm{FoM}_1 \times \left(\frac{\mathrm{TC}}{T_{Range}} \right) \tag{24}$$

where $T_{Range} = T_{Max} - T_{Min}$. Equation (24) is identical to the composite FoM introduced in [23]. Fig. 19(a) and (b) visualizes the benchmark comparisons in terms of LS versus V_{DD}, Min and TC/T_{Range} versus FoM_1, respectively.

In resistor-based current references, the silicon area, supply voltage, and current are bound by a fundamental tradeoff.

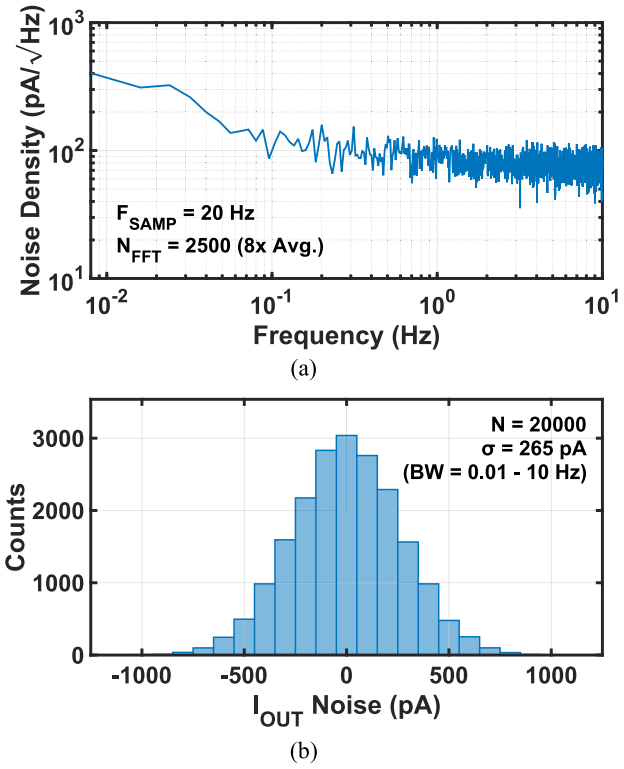
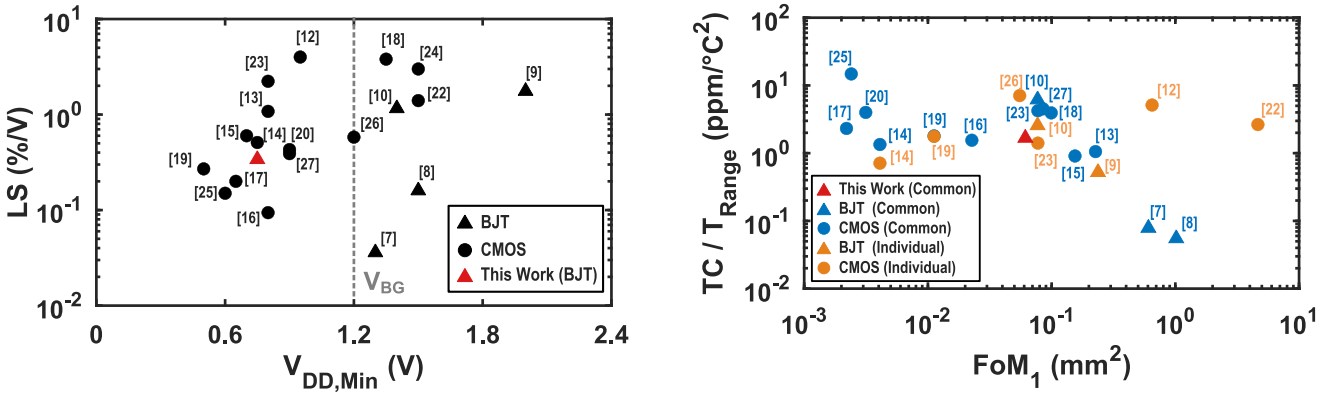


Fig. 18. Measured I_{OUT} noise characterization. (a) Noise spectrum and (b) histogram.

Traditional BJT-based references attain the strongest process immunity, but only through a large basis voltage that requires large resistors and area, resulting in an unfavorable power–area cost. In contrast, CMOS-based references reduce the supply voltage and silicon area by forming the basis from ΔV_{TH},

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(a) (b)

Fig. 19. Design benchmarks with recently reported state-of-the-art current references. (a) LS versus minimum supply voltage and (b) TC/T_{Range} versus FoM_1, where the product of the two coordinates gives FoM_2.

but this basis is itself strongly process-dependent, sacrificing process robustness. The proposed β_R-based extraction lies outside this tradeoff: its millivolt-level basis enables operation from a 0.75-V supply without relying on the bandgap voltage, achieving the lowest FoM_1 among the BJT-based references with a sub-megaohm resistor and a compact area. Under FoM_2, which additionally weights temperature stability, the ranking within the BJT-based group changes: the microwatt-class designs [7], [8] lead, reflecting the multi-point compensation their power budget affords, while the proposed reference remains the lowest among the nW-class BJT references. Within this class, the measured 277 ppm/°C is obtained with a single common code, whereas the common-code TC of [10] is 680 ppm/°C; 89 ppm/°C of [9] is lower, but is reached with per-chip trimming, and the 60-MΩ resistor and 2-V supply it requires place its FoM_2 above that of this work. Beyond these figures of merit, the simulated process-induced variation of the proposed reference lies between that of bandgap and CMOS-based references, combining a lower process sensitivity than CMOS-based bases with the compact, sub-1 V efficiency that bandgap-based bases cannot offer.

V. Conclusion

This article presented a low-power current reference based on a new physical basis, $V_{\beta R}$, derived from the reverse current gain (β_R) of a saturation-region BJT. By exploiting the self-cascode interaction of two identical n-p-n BJTs, the proposed mechanism extracts a millivolt-level reference voltage determined entirely by intrinsic bipolar device parameters. Analytical formulation establishes that $V_{\beta R}$ simultaneously achieves a magnitude approximately 150× smaller than the conventional bandgap voltage and a moderate process sensitivity, positioned between conventional BJT and CMOS bases. A current reference circuit implementing this principle was fabricated in a standard 0.18-μm CMOS process, occupying 0.025 mm^2 and consuming 30 nW from a 0.75-V supply. Measurements across ten samples without individual-code trimming demonstrate a TC of 277 ppm/°C over −40 °C to 125 °C and an LS of 0.34%/V over 0.75–2.0 V. The proposed β_R-based extraction establishes that a smaller process spread than that of CMOS-based bases can be retained at a significantly reduced voltage magnitude, making it a viable reference basis for severely power- and area-constrained IoT systems.

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